
Applied Cyber Threat Analysis – Windows Log Analysis

- This lab was conducted within the Virginia Cyber Range Environment.
 - Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing
 - Cheche Agada. (2024). Email Header Analysis Lab IT-462: Applied Cyber Threat Analysis, George Mason University
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Introduction:

Logs are an important tool for security monitoring and investigations. All operating systems and many software applications maintain some kind of log. Such logs record system activities in real time and are useful for troubleshooting, tracking user activity, and monitoring system performance. The Windows Event Viewer is a tool built into Windows to access and view several logs maintained by the operating system.

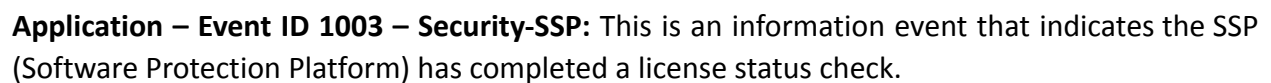
Objective:

View Windows logs and interpret events using Windows Event Viewer.

Task 1 – Identifying Windows Events

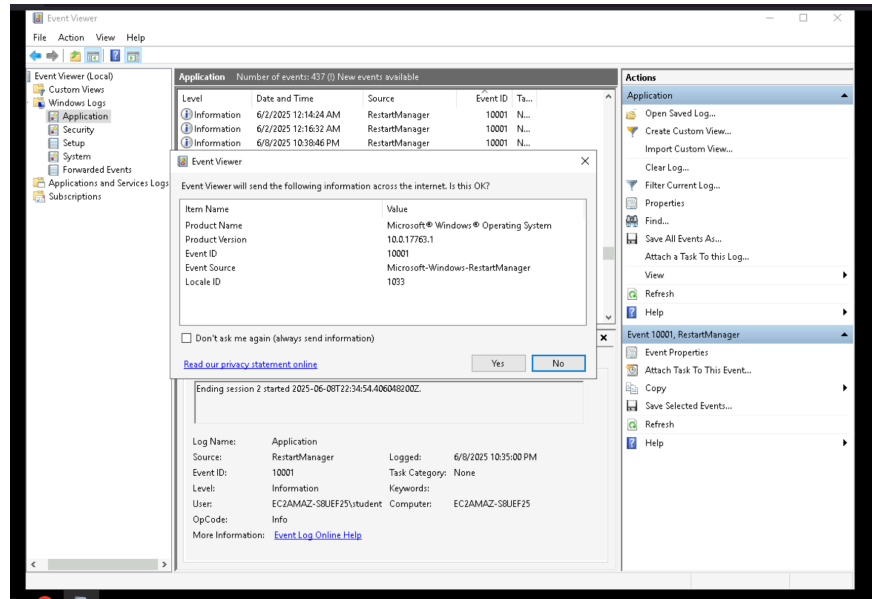
Using the Windows Event Viewer, Identify at least 10 different events in the Security, System, or Application Log(s). Research these events and write short notes on each one. Explain what the event is, why it occurred, and what effect (if any) it has on the system.

- Use event IDs to differentiate one event from another. Do not discuss one type of event more than once.



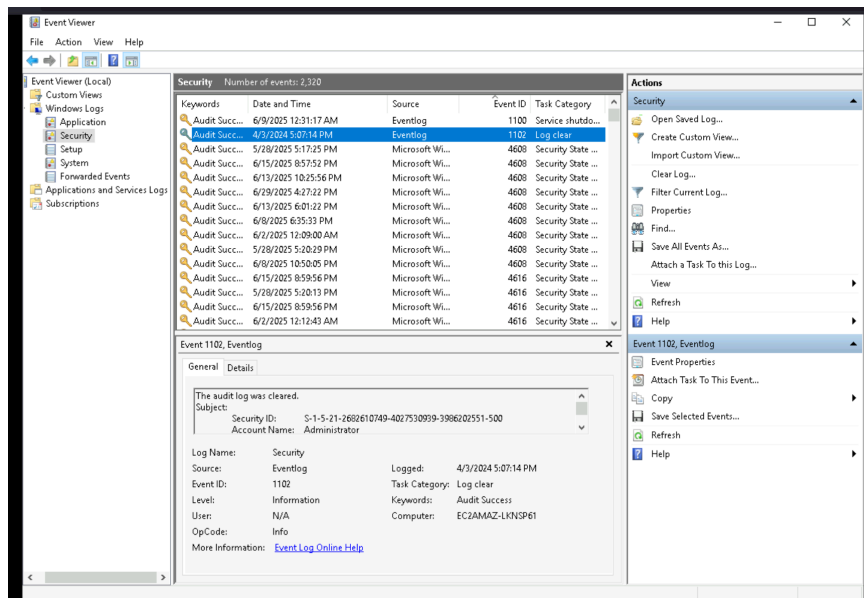
Troy Fracyon
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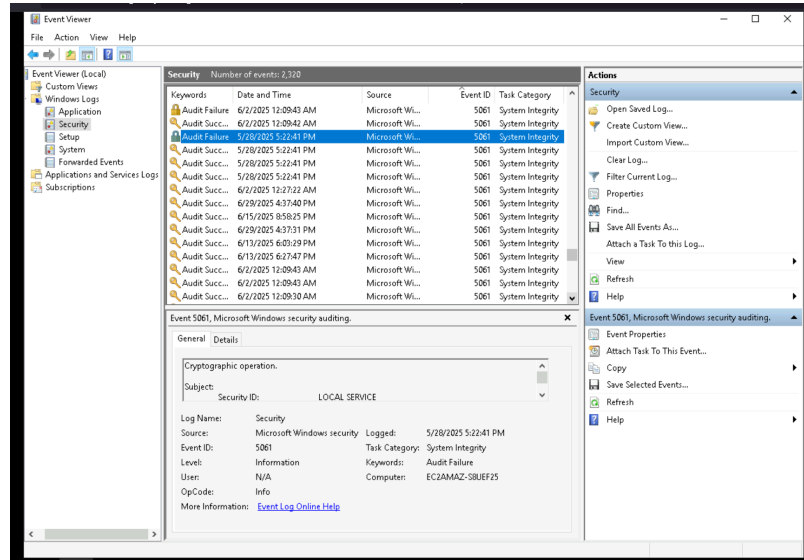
Application – Event ID 10001 – RestartManager: This event indicates that a session has ended successfully.

4.



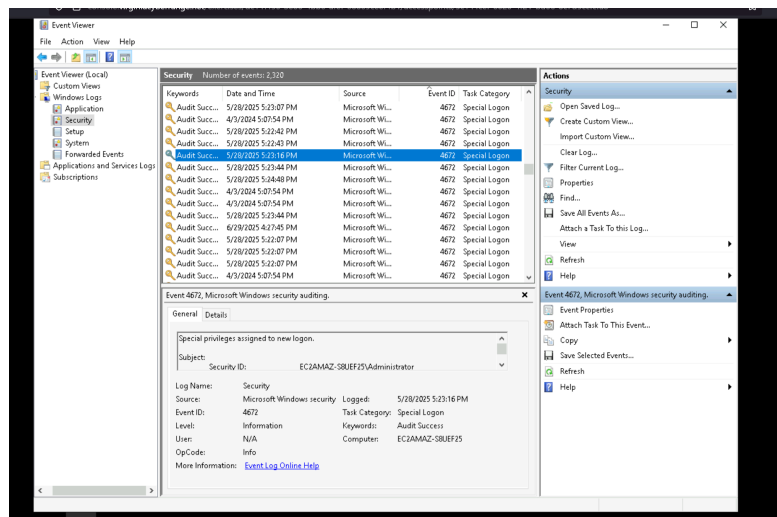
Security – Event ID 1102 – Eventlog: This event indicated that the event log was cleared. Security log clears should only be done by the administrator

5.



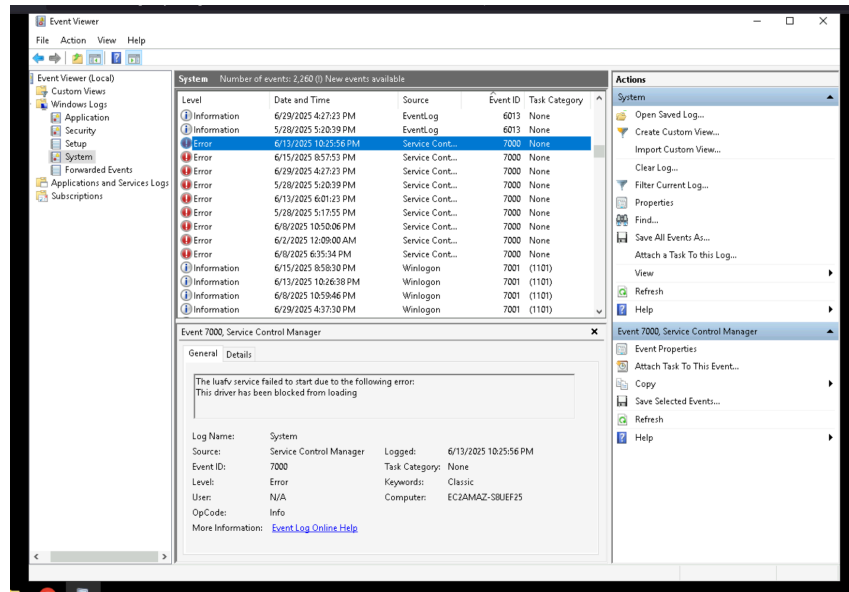
Security – Event ID 5061 – System Integrity (Audit Failure): There are a few accompanying messages with the event log 5061. This event log relates to cryptographic operations within windows. When there is an Audit failure, there is an issue with the key storage.

6.



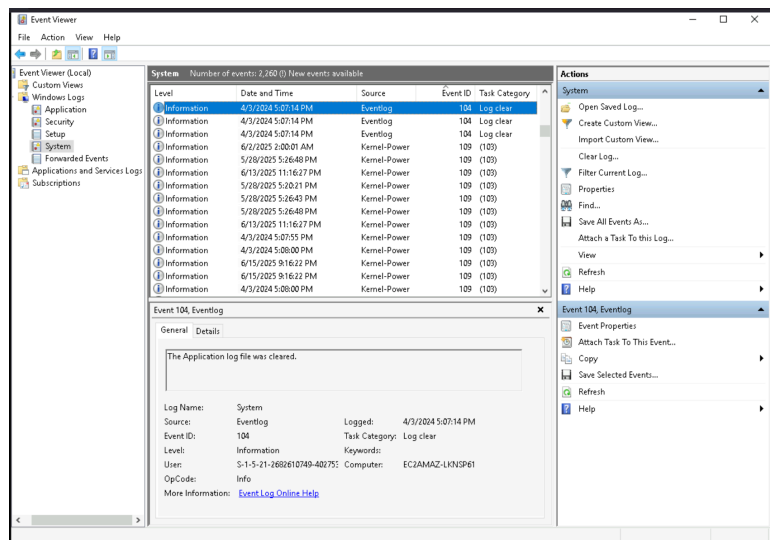
Security – Event ID 4672 – Special Logon: This event log indicates that special privileges were granted/assigned to a new session.

7.



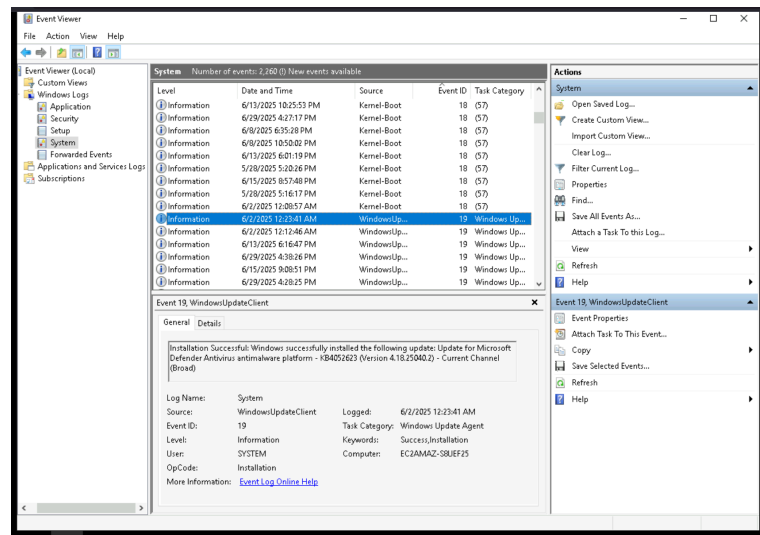
System – Event ID 7000 – Service Control Manager (Error): This event log is indicating that a service failed to start (luafv driver in this circumstance). This can be due to config, dependency, or hardware issues.

8.



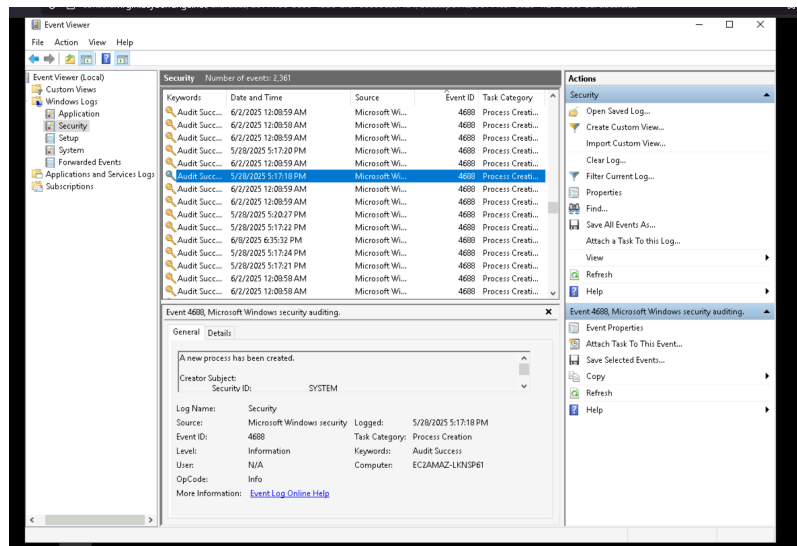
System – Event ID 104 – Eventlog (log clear): There were 3 log clears in this event log for Application, System log, and Setup log. Most likely for maintenance procedures.

9.



System – Event ID 19 – WindowsUpdateClient(Agent): This event log is indicating a successful update for Microsoft Defender Antivirus antimalware platform.

10.



Security – Event ID 4688 – Process Creation: This event log is used to indicate that a process was created, the information collected includes the user who created the process, the process ID and the command line used.